

Lecture 4: Tetanus, botulism, prions

Tetanus

Tetanus, also known as lockjaw, is a serious disease that attacks the nervous system with the potentially deadly bacteria called *Clostridium tetani*. It can cause painful spasms and stiffness in our jaw muscles. These spasms usually last a few minutes each time and occur frequently for three to four weeks. Spasms may be so severe that bone fractures may occur.

Clostridium tetani: Gram-positive, motile bacteria. Endospores-forming bacteria: drum-stick appearance due to terminal round spore with a diameter greater than vegetative cell. Rod-shaped, named after Greek word for spindle, *kloster*. Obligate anaerobes: because they lack enzymes necessary for neutralizing oxygen radicals as : (Superoxide Dismutase (SOD) and Catalase)



Virulence factors: Production of two exotoxins

- 1- Tetanolysin: lyse RBCs with no function determined
- 2- Tetanospasim: Neurotoxin, one of most potent toxins known: minimum human lethal dose: 2.5 ng/kg body weight, under control of plasmid gene. Vegetative cells produce tetanospasmin during the stationary phase and release it mainly when they lyse, Heat-labile (destroyed at 65°C) and destroyed by intestinal proteases, Toxic to man and various animals when injected parenterally, but it is not toxic by the oral route.

Tetanospasmin targets and actions:

Targets: Several sites within the CNS, including spinal cord and brain, Peripheral nervous system (PNS): Autonomic nervous system (ANS) and Somatic nervous system (SNS)

Actions: Absorption into CNS, absorbed into axon and transported across synaptic junctions until reaches CNS, Circulates through circulatory and lymphatic systems, Rapidly fixed to gangliosides at the presynaptic junction of inhibitory motor nerve endings

Mode of Action: Blocks inhibitory impulses via interference with inhibitory neurotransmitter release, including that of glycine and gamma-amino butyric acid and causes unopposed muscle contraction and spasm and seizures

Mode of Infection: C. tetani is found worldwide. Ubiquitous in soil, and occasionally found in intestinal flora of humans and animals.

A. tetani is the cause of tetanus, or lockjaw. when spores are introduced into wounds by contaminated soil or foreign objects such as dirty nails or glass splinters, knives, tools, wood splinters, and animal bites. Women face an additional risk of infection if a contaminated tool is used during childbirth or during an abortion. Tetanus is not transmitted from person to person.

Clinical Picture: The onset of symptoms is typically three to twenty-one days following infection.

Fever, sweating, headache, trouble swallowing, high blood pressure, and a fast heart rate. It may take months to recover. About ten percent of cases prove fatal.

Clinical forms:

Generalized:Involvement of bulbar and paraspinal muscles (trismus or lockjaw, risus sardonicus, difficulty swallowing, irritability, opisthotonos); involvement of autonomic nervous system (sweating, hyperthermia, cardiac arrhythmias, fluctuations in blood pressure)

Cephalic: Primary infection in head, particularly ear; isolated or combined involvement of cranial nerves, particularly seventh cranial nerve; very poor prognosis

Localized: Involvement of muscles in area of primary injury; infection may precede generalized disease; favourable prognosis

Neonatal: Generalized disease in neonates; infection typically originates from umbilical stump very poor prognosis in infants whose mothers are non-immune
 Diagnosis: primarily a clinical diagnosis; organism is rarely isolated.

Treatment: 3 main targets: 1. Control of muscle spasms: Muscle relaxants. 2. Halting of toxin production: Metronidazole and intramuscular penicillin 3. Neutralization of toxin effects (Hyperimmune human globulin (TIG) to neutralize toxin). TIG: binds to and eliminates unbound toxins from body but cannot affect already bound toxins, as this is an irreversible event

Prevention: Rigorous hygienic response to injury (wound management).

Vaccination: First 4 immunization shots (DTP: diphtheria-tetanus-pertussis) given within 2 years. Every 10 years: booster shot

Wound Management:

Patient	Not Tetanus Prone	Tetanus Prone
	Linear, 1 cm deep cut, without devitalized tissue, without major contaminants, less than 6 hours old	Blunt/missile, burn, frostbite, 1 cm deep; devitalized tissue present + contaminants (e.g., dirt, saliva); any wound 6 hours old
Not completed primary or vaccination history unknown	Vaccine	Vaccine and TIG*
Completed primary series	Vaccine if more than 10 years since last booster	Vaccine if more than 5 years since last booster

Botulism

A rare and potentially fatal illness caused by a toxin produced by the bacterium *Clostridium botulinum*. The disease begins with weakness, blurred vision, feeling tired, and trouble speaking. Botulism: (Revise Food-poisoning lecture in GIT block). Causative organism: *C. botulinum*.

Virulence: Botulinum toxin (A-B polypeptide neurotoxin (actually a series of 7 antigenically different; type A and B most common), Coded for by a prophage (lysogenized *Clostridium botulinum*), Highly toxic, Heat labile (unlike staph), 10 minutes 60.0°C.

Mechanism of action: Absorbed by gut and carried by blood to peripheral nerve synapses. Blocks release of acetylcholine at the myoneuronal junction resulting in a reversible flaccid paralysis

Transmission: Eating preserved exotoxin in improperly canned food and honey contaminated with spores

Types of Botulism:

Disease	Adult	Infant
Acquisition	Preformed toxin ingested (toxicosis) Poorly canned alkaline vegetables (green beans), smoked fish	Spores ingested: household dust, honey Toxin produced in gut (toxi-infection)
Symptoms	1–2 day onset of weakness, dizziness, blurred vision, flaccid paralysis (reversible); $\pm$ diarrhea, nausea or vomiting	Constipation, limpness/flaccid paralysis (reversible): diplopia, dysphagia, weak feeding/crying; may lead to respiratory arrest
Toxin demonstrated in	Suspected food or serum	Stool or serum
Treatment	Respiratory support Trivalent (A-B-E) antitoxin	Respiratory support in monitored intensive care; hyperimmune human serum Antibiotics generally not used as may worsen or prolong
Prevention	Proper canning; heat all canned foods	No honey first year

Prions

The actual infectious agent consists merely of protein and is capable of replicating and transmitting infections without the need for informational nucleic acids.

Prions are atypical agents:

SIMILAR TO VIRUSES

- small
- filterable
- need host cells
- no machinery for energy generation or protein synthesis

DIFFERENT FROM VIRUSES

- no detectable virions in infected tissues
- no detectable virions in purified infectious material
- if nucleic acid is present, very small
- very resistant to inactivation

Prions are proteins found in normal cells but, in genetically predisposed individuals they transform into a pathogenic abnormal form. Prions resist heat and chemicals, including formaldehyde. The infectious agent infects a range of mammals, including humans, scrapie in sheep and cattle, and replicates slowly

with a long incubation period (SLOW Infections). Prions are the only known infectious pathogens that are devoid of nucleic acid. Prion disease may be manifest as infectious, genetic and sporadic disorders; no other group of illnesses with a single etiology presents with such a wide spectrum of clinical manifestations. Prions disease result from the accumulation of PrP^{Sc}. the confirmation of which differs substantially from that of its precursor PrP^C. PrP^{Sc} can exist in a variety of different conformations, each of which seems to specify a particular disease phenotype.

Diagnosis is based on histological changes in the brain.

Different prion Diseases

Disease	Infectious agent	Host	Comments
Kuru	Prion	Human	Subacute spongiform encephalopathy (SSE); Fore Tribe, New Guinea; cannibalism
Creutzfeldt-Jakob disease (and variant)	Prion	Human	SSE Genetic predisposition; ingestion of infected cow brains
Gerstmann-Sträussler-Scheinker	Prion	Human	SSE
Fatal familial insomnia	Prion	Human	SSE
Scrapie	Prion	Sheep	SSE—scraping their wool off on fences

Creutzfeldt–Jakob disease (CJD): This is a rare encephalopathy of humans with an uncertain mode of transmission. Transmission has been associated with corneal grafts, neuro- surgery and human growth hormone preparations. Variant CJD (vCJD), first reported in the UK, is thought to have resulted from transmission of infection from BSE in cattle to humans via contaminated beef. Prions are not destroyed by normal sterilization cycles in autoclaves; killing requires 134 C for 18 min or exposure to sodium hydroxide.

Kuru: Transmission is from human to human by cannibalism and recorded mainly in Papua New Guinea.

Suggested Treatment: Interfering with the conversion of PrP^C into PrP^{Sc} would seem to be the most attractive therapeutic target. One reasonable therapeutic

strategy would be to stabilize the structure of PrPC by binding a drug. Alternatively, drugs that destabilize the structure of PrPSc might also prove useful.